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Key Points:

- ICME-driven activity peaks during solar maxima, and HSS-driven activity dominates the declining and minimum phases
- High-speed stream events show a pronounced pre-midnight hotspot, consistent with substorm-driven activity
- ICME-driven storms activate multiple current systems across a broader Magnetic Local Time range, including post-midnight enhancements

Supporting Information:

Supporting Information may be found in the online version of this article.

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On the Effects of the Solar Wind Structure in the Global Distribution of dB_H/dt Spikes During Geomagnetic Storms

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Abstract Variations in the solar wind, particularly those associated with interplanetary coronal mass ejections (ICMEs) and high-speed streams (HSSs), drive geomagnetic disturbances that produce rapid changes in the ground magnetic field. These variations, commonly quantified by the time derivative of the horizontal magnetic field (dB_H/dt), are closely related to geomagnetically induced currents (GICs), which pose risks to technological infrastructure. In this study, we analyze 609 geomagnetic storms ($SYM-H \leq -50$ nT) from 1995 to 2024, interplanetary coronal mass ejection and HSS driven events, to investigate how different solar wind drivers modulate the occurrence, intensity, and spatiotemporal distribution of dB_H/dt spikes. We find that dB_H/dt enhancements are concentrated between 60° and 75° magnetic latitude for both drivers, but exhibit distinct Magnetic Local Time (MLT) distributions. HSS-driven storms predominantly produce pre-midnight enhancements, consistent with substorm-related processes, whereas ICME-driven storms generate a broader MLT distribution, including significant post-midnight activity associated with multiple current systems, such as the substorm current wedge and field-aligned currents. Additionally, we show that the occurrence and intensity of dB_H/dt spikes vary systematically with the solar cycle, with ICME-driven activity peaking during solar maxima and HSS-driven activity dominating the declining and minimum phases. These results provide a unified long-term perspective on how solar wind drivers control rapid geomagnetic field variations, highlighting the importance of distinguishing between driver types for improving space weather modeling and GIC risk assessment.

Plain Language Summary The space around Earth is influenced by the solar wind, a stream of charged particles emitted by the Sun. At times, this flow intensifies due to events such as coronal mass ejections (large eruptions of solar material) or high-speed streams (fast outflow solar wind). These disturbances can cause rapid changes in Earth's magnetic field, measured as the rate of change of the horizontal magnetic field (dB_H/dt). These changes can generate electric currents in the ground, known as geomagnetically induced currents, which may affect power grids, pipelines, and railways. In this study, we analyzed more than 600 geomagnetic storms that occurred between 1995 and 2024. We compared storms driven by different types of solar wind and found that each type produces distinct patterns in when and where magnetic disturbances occur. We also show that these effects vary with the 11-year solar activity cycle. These results improve our understanding of how solar wind conditions affect Earth's magnetic environment and can help enhance forecasting models used to protect technological systems from space weather impacts.

1. Introduction

Solar transient events, such as interplanetary coronal mass ejections (ICMEs) and high-speed streams (HSS), when directed toward Earth, interact with the Earth's magnetosphere and can cause strong perturbations in this environment, known as geomagnetic storms (Echer et al., 2006; Gonzalez et al., 1999). These different solar wind structures drive geomagnetic activity through distinct physical mechanisms and produce markedly different magnetospheric responses, including variations in storm intensity, duration, and energy coupling efficiency (Borovsky & Denton, 2006; Denton et al., 2006). The intensity of a geomagnetic storm is primarily dictated by the strengthening of the ring current, which perturbs the global surface magnetic field. Indices such as the hourly

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Disturbance Storm-Time (Dst) index and the higher-resolution SYM-H index are used to quantify the strength of this current by measuring the magnitude of this magnetic perturbation (Sugiura, 1965). Geomagnetic storms typically progress through distinct phases. The initial phase is often marked by a sudden positive excursion in the SYM-H (or Dst) index, known as a sudden storm commencement. This signature is driven by the dynamic pressure enhancement of the solar wind, characteristic of interplanetary coronal mass ejection (ICME) fronts, resulting in the compression of the day-side magnetopause and the intensification of the Chapman-Ferraro currents (Chapman & Ferraro, 1931; Dessler et al., 1960). This is followed by a main phase, when the ring current intensifies, and a recovery phase, as the ring current returns to typical values (Kamide & Chian, 2007). Rapid changes in geomagnetic field intensity during these storm phases directly affect electrical currents in the magnetosphere and ionosphere, leading to fluctuations measured at ground-based magnetometers as sharp temporal variations in the magnetic field (dB/dt). Here, dB/dt is used in a general sense to describe rapid magnetic field variations without restriction to a specific component. Such variations can induce currents on ground-based conductive infrastructure, known as geomagnetically induced currents (GICs) (Gil et al., 2021; Smith et al., 2022). Hence, understanding the modulation of geomagnetic field perturbations by different solar wind structures is crucial both for scientific insights and practical space weather mitigation efforts.

Extensive research has been conducted into the spatiotemporal characteristics of these perturbations. Instances of large dB/dt have been shown to occur in three main local time hot-spots: the pre-midnight, dawn, and pre-noon sectors (Juusola et al., 2015; Kataoka & Pulkkinen, 2008; Milan et al., 2023; Schillings et al., 2022; Viljanen et al., 2001; Weigel et al., 2003). Complementing these observations, Viljanen and Tanskanen (2011), using high-latitude magnetometer data, observed intense activity in the horizontal component of the geomagnetic field perturbations (dB_H/dt) predominantly occurring around midnight and early morning, diminishing significantly by noon. Analyses of extreme geomagnetic fluctuations indicate that the magnitude and occurrence probability of dB_H/dt are controlled by latitude-dependent physical drivers, with the 99.97th percentile peaking near 67° Corrected Geomagnetic coordinate (CGM), while the 100-year return levels shift equatorward to approximately 53° (Rogers, Wild, Eastoe, Gjerloev, & Thomson, 2020). These fluctuations are primarily driven by westward electrojet intensifications during substorm onsets in the pre-midnight sector, 20–24 Magnetic Local Time (MLT), whereas sustained variations in the dawn sector, 03–10 MLT, are dominated by ultralow frequency (ULF) wave power and Pc5 pulsations. At low-to-mid latitudes, the extreme tail of the distribution is instead controlled by magnetospheric compressions from interplanetary shocks, resulting in sudden commencements that peak near local noon (Rogers, Wild, & Eastoe, 2020; Rogers et al., 2021). More recent global observational efforts by Schillings et al. (2022) utilized global magnetometer networks to show that magnetic spikes during geomagnetic storms exhibit distinct behavior, progressing from the pre-midnight sector toward the morning sector. Interestingly, their analysis revealed no direct correlation between spike frequency and storm intensity (as measured by the disturbance storm time (Dst) or auroral electrojet (AE) indices), underscoring the complexity of factors influencing GICs risk.

Recent work has demonstrated that rapid magnetic field variations are closely linked to magnetosphere–ionosphere coupling processes, particularly the dynamics of field-aligned currents (FACs) and substorm activity (A. Fleetham et al., 2024). These studies show that dB/dt spikes frequently occur in association with enhanced FAC activity and are co-located with regions of intensified ionospheric and FAC systems, highlighting the importance of current system evolution in shaping ground magnetic perturbations.

Different current systems are responsible for generating dB_H/dt spikes at different local times. Milan et al. (2023) identified three primary hotspots and linked them to specific physical phenomena. In the pre-midnight sector, activity is associated with substorm onset and the formation of the Substorm Current Wedge (Apatenkov et al., 2020; Juusola et al., 2015; Kataoka & Pulkkinen, 2008; Ngwira et al., 2018; Weigel et al., 2003; Zou et al., 2022), typically manifesting in ground magnetometer data as a sudden negative excursion in the north–south component of the geomagnetic field. In the dawn sector, spikes are linked to omega bands and associated Ps6 ULF pulsations (Rajaram et al., 1986; Rostoker et al., 1980), which have periods in the 10–20 min range and appear in both the north–south and east–west components of the geomagnetic field (Milan et al., 2023). Finally, in the pre-noon sector, activity is associated with Kelvin–Helmholtz instability (KHI) at the magnetopause (Engebretson et al., 2020). Early work using high-resolution magnetometer data, from the International Monitor for Auroral Geomagnetic Effect (IMAGE) network, demonstrated that this daytime peak is driven primarily by solar wind velocity rather than by the interplanetary magnetic field (IMF) B_z , with these perturbations taking the form of ~5 min ULF pulsations observed in all three components of the ground magnetic field.

Beyond these specific local time phenomena, other factors in the upstream solar wind are critical. While the role of IMF B_z is well-established, the influence of other components remains less explored. Eastwood et al. (2005) highlighted that a predominant IMF B_x could generate turbulence in the upstream foreshock region, subsequently producing ULF waves capable of penetrating and disturbing the magnetosphere-ionosphere-ground system. Using explainable techniques in machine learning models, Coughlan et al. (2025) showed asymmetries in the impact of the solar wind B_x component on the dawn and dusk occurrence of ground magnetic field perturbations. We note, however, that the IMF B_y component has also been shown to play a significant role in modulating high-latitude geomagnetic activity, including a pronounced seasonal dependence and polarity-driven asymmetries (Holappa & Mursula, 2018). Furthermore, the nature of the solar wind driver itself dictates the perturbation patterns. It is well established that ICME- and HSS/CIR-driven storms exhibit fundamentally different magnetospheric characteristics, including differences in plasma sheet properties, ring current development, and temporal evolution (Borovsky & Denton, 2006; Denton & Borovsky, 2012; Denton et al., 2006). In particular, ICME-driven storms tend to be more intense and impulsive, whereas HSS/CIR-driven storms, where geomagnetic activity is often initiated by the turbulent, high-field Corotating Interaction Region (CIR) and sustained by the subsequent HSS, are typically more prolonged and moderate. Pedersen et al. (2024), using high-resolution IMAGE magnetometer network data, demonstrated notable variations in spike occurrence and distribution depending on the type of solar wind driver, emphasizing distinct spike patterns during sheath-driven, magnetic cloud-driven, and HSS-driven storms. This provides critical insights into storm-time perturbation dynamics.

While previous studies have investigated dB_H/dt characteristics and their association with geomagnetic activity, including detailed analyses of spike occurrence, current systems, and storm-time dynamics (A. Fleetham et al., 2024; Milan et al., 2023; Schillings et al., 2022), a unified framework that simultaneously accounts for solar wind driver type, storm phase, and long-term variability in geomagnetic activity is still lacking.

This work addresses these gaps by providing an unprecedented global statistical analysis spanning nearly three decades (1995–2024), which extends coverage to integrate data from the highly active peak of the current Solar Cycle 25. The core novelty of this study lies in utilizing the extensive SuperMAG network to construct a high-resolution mapping of dB_H/dt spikes into fine-scale MLT and latitudinal bins, while explicitly separating ICME- and HSS-driven storms across their individual initial, main, and recovery phases. By doing so, we advance beyond bulk statistical averages to identify distinct, phase-specific current system activations and cycle-dependent hazard profiles, establishing a critical baseline for driver-targeted space weather mitigation.

2. Data Set and Methods

The ground magnetic perturbations were collected from the SuperMAG global consortium of magnetometer networks (Gjerloev, 2012). We used 1-min-resolution data from all available stations at mid- and high magnetic latitudes ($50^\circ \leq \text{MLAT} \leq 90^\circ$). The use of 1-min resolution data is consistent with standard geomagnetic observatory products and allows for robust global statistical analysis over multi-decadal timescales. While higher-cadence measurements can capture fine-scale and rapid fluctuations, the dominant variations associated with storm-time processes and large-scale dB_H/dt enhancements are well represented at 1-min cadence (Rogers et al., 2021). The coordinate system provided by SuperMAG is NEZ, where N is positive in the direction of the local magnetic north, E is positive in the local magnetic east direction, and Z is positive in the vertical downward direction. For this analysis, we are interested in the global measurements of the horizontal component of the geomagnetic field since 1995; therefore, the minute variation, or dB/dt for the horizontal component, was estimated using finite differences between consecutive samples as

$$dB_H/dt = \sqrt{(\Delta B_N/\Delta t)^2 + (\Delta B_E/\Delta t)^2},$$

where ΔB_N and ΔB_E represent the differences between consecutive measurements and $\Delta t = 1$ min. Figure 1 shows the global coverage of the SuperMAG magnetic stations used in this work. Although the geographic distribution of these stations exhibits longitudinal gaps, particularly over oceanic regions, its impact on the MLT analysis is significantly minimized by two main factors. First, the extensive temporal coverage of nearly three decades ensures that stations continuously sample all MLT sectors due to the daily rotation of the Earth. Second, our statistical analysis utilizes relative occurrence frequencies within specific MLT-latitude bins rather than

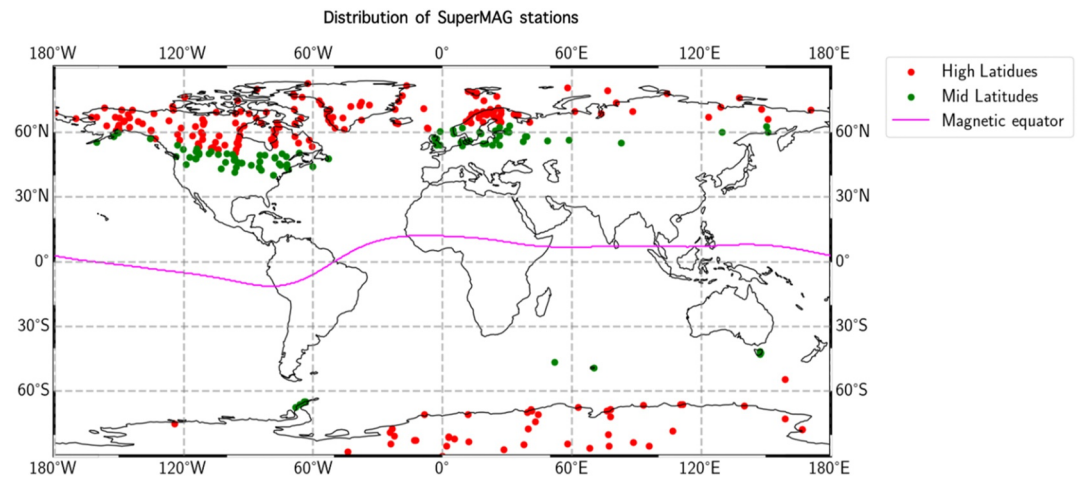


Figure 1. Distribution of SuperMAG magnetic station location used in this work. The red points are located at high magnetic latitudes, and the green dots represent mid-latitude stations. The magenta line represents the geomagnetic equator.

absolute counts, an approach that effectively normalizes the uneven spatial density of the ground network (Bergin et al., 2020; Newell & Gjerloev, 2012).

The ground magnetic field data were selected over a time frame corresponding to the minimum of the SYM-H index. The algorithm to identify and separate the storms was constructed based on the following criteria, illustrated in Figure 2: a storm is first identified when the index reaches -50 nT or less. The minimum of SYM-H marks the beginning of the recovery phase. The recovery phase ends when the index returns to intensities 25% of the minimum SYM-H, or quiet levels of -15 nT. The main phase begins where the SYM-H changes sign before the minimum point. To define the start of an initial phase, simply find the point where the index reaches quiet levels before reaching its local maximum. This list is an extension of the list used in Pinto et al. (2022), covering 29 years of data from 1995 to 2024. The list comprises events with weak intensities through super-storm events (Gonzalez et al., 1994).

We separated the storms based on the solar wind structure, which could be related to the perturbation. For this study, we classified perturbations associated with ICMEs and HSSs. The structures were identified based on the publicly available lists and catalogs, such as the Richardson Lists, available at <https://izw1.caltech.edu/ACE/ASC/DATA/level3/icmetable2.htm> (Cane & Richardson, 2003), and the Space Weather Database Of Notifications, Knowledge, Information (DONKI), available at <https://kauai.ccmc.gsfc.nasa.gov/DONKI/search/>. We also compared with other published lists of interplanetary events (Da Silva et al., 2021, 2023; Gabrielse et al., 2019; Juusola et al., 2023; Marchezi et al., 2022; Pedersen et al., 2024; Walach & Grocott, 2019), as well as visual analysis of solar wind parameter data from the NASA/GSFC's OMNI data set through OMNIWeb (<http://omniweb.b.gsfc.nasa.gov>). A total of 609 events were selected, comprising 304 ICME-driven and 305 HSS-driven storms (Marchezi & Pinto, 2025).

Figure 2 compares the geomagnetic responses during an ICME event (left) occurring between 17 and 19 March 2015, and a HSS event (right) from 3–7 July 2015. These events were selected as representative examples to illustrate the distinct geomagnetic signatures and temporal evolution associated with ICME- and HSS-driven storms, and are not intended to represent the full statistical data set analyzed in this study. The top panels display the geomagnetic disturbance quantified by the SYM-H index. The lower panels show the 1-min variation of the horizontal magnetic field component (dB_H/dt) recorded at three ground magnetometer stations situated at different geomagnetic latitudes: Gjoa Haven (77.5° MLAT), Puvirnituq (69.6° MLAT), and Ottawa (55.0° MLAT). The ICME event triggered a sudden and intense geomagnetic storm, characterized by a sharp decrease in SYM-H to nearly -230 nT, followed by a relatively slow recovery phase. In contrast, the HSS event produced a more gradual and less intense storm, with a minimum SYM-H around -100 nT, but sustained activity over several days.

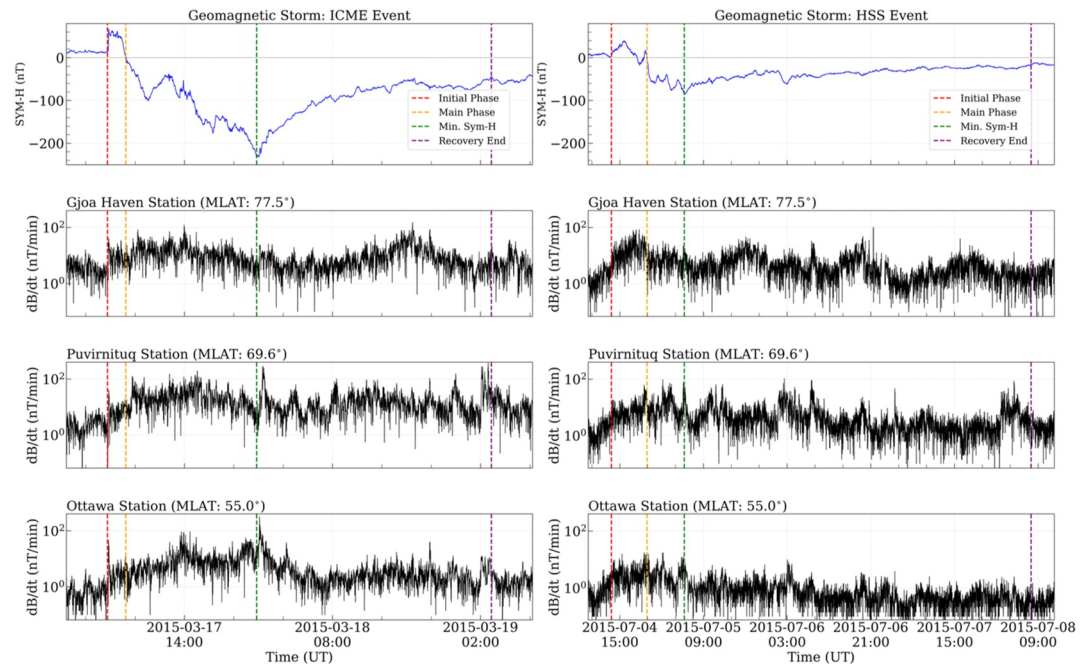


Figure 2. Comparison of geomagnetic responses to two geomagnetic storms driven by different solar wind structures: an interplanetary coronal mass ejection (ICME, left panels) and a high-speed solar wind stream (high-speed stream (HSS), right panels). The top panels show the global SYM-H index, with vertical dashed lines marking key storm phases: the onset of the initial and main phases, the minimum SYM-H, and the end of the recovery phase. Bottom panels display the time derivatives of the magnetic field (dB_H/dt), in logarithmic scale, measured at three ground magnetometer stations at different magnetic latitudes: Gjoa Haven (77.5° MLAT), Puvirnituq (69.6° MLAT), and Ottawa (55.0° MLAT). The ICME event (March 2015) produced a more intense and abrupt storm, with sharp dB_H/dt spikes, while the HSS event (July 2015) generated prolonged, lower-magnitude fluctuations, especially at higher latitudes. The same y-axis ranges are used for both ICME and HSS panels to enable direct comparison of magnitudes between the two event types.

During the ICME storm, large-amplitude and short-duration spikes in dB_H/dt are evident at higher latitude stations (i.e., Gjoa Haven and Puvirnituq), particularly around the onset and main phase of the storm. The most intense fluctuations were observed at those stations, consistent with the enhanced ionospheric current systems that occur during strong geomagnetic activity. The HSS-driven storm exhibited a higher frequency of moderate dB_H/dt enhancements, especially at Gjoa Haven and Puvirnituq, but with lower peak magnitudes than the ICME case. The Ottawa station recorded substantially weaker responses, specially during the HSS event, with around one order of magnitude reduction, reflecting the reduced sensitivity of mid-latitude locations to auroral processes during those events. While ICME storms more often pose a greater immediate threat due to their abrupt and intense magnetic field variations, HSS events can induce prolonged periods of moderate geomagnetic activity that may contribute to cumulative GIC effects (Pulkkinen et al., 2008), particularly at high latitudes, with different solar wind drivers (e.g., ICMEs and the combined HSS/CIR structures) producing distinct geomagnetic responses (Richardson et al., 2006).

These differences indicate that geomagnetic hazard depends not only on storm intensity but also on its temporal evolution and latitudinal impact. ICME-driven events tend to produce short-lived but extreme perturbations, whereas HSS-driven events generate prolonged moderate activity that can lead to cumulative effects, particularly at high latitudes.

In cases where storms exhibit intervals during the recovery phase in which the SYM-H index shows renewed intensification, these enhancements were operationally classified as new storms only when a further decrease in SYM-H (i.e., intensification toward more negative values) reached at least 50 nT relative to the preceding local minimum. In such cases, the recovery phase of the preceding major storm is considered to terminate with the initiation of the subsequent storm. In the following sections, we conduct a statistical evaluation of these events to characterize their occurrence and underlying features.

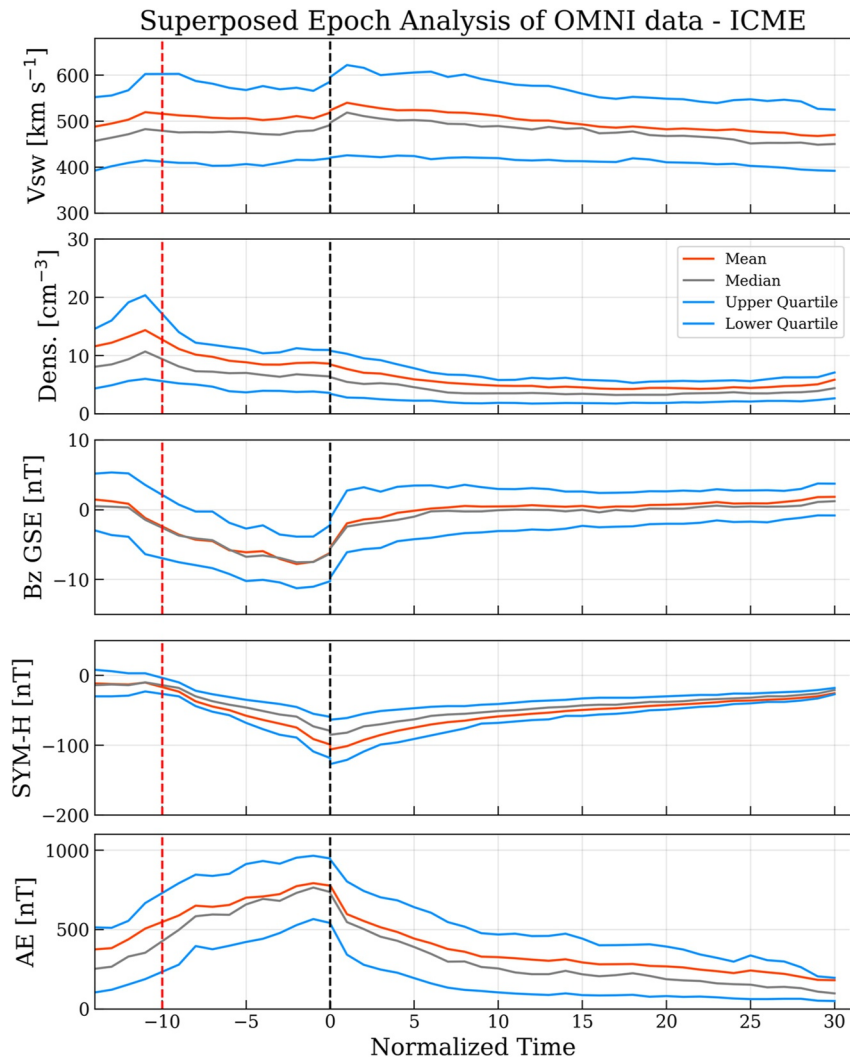


Figure 3. Time-normalized superposed epoch analysis of solar wind and geomagnetic parameters during interplanetary coronal mass ejection (ICME) events. From top to bottom: solar wind speed, proton density, interplanetary magnetic field B_z component, SYM-H index, and auroral electrojet index. The red line represents the mean, the gray line the median, and the blue lines indicate the upper and lower quartiles, illustrating the variability across events. The same y-axis ranges are used for both ICME and high-speed stream panels to enable direct comparison of magnitudes between the two event types.

3. HSS Versus ICME Dependence of dB_H/dt

Figures 3 and 4 present the results of a superposed epoch analysis of solar wind and geomagnetic parameters during ICME and HSS events, respectively. The analysis employs a normalized time axis, where the epoch (time zero, indicated by the vertical dashed black line) corresponds to the minimum of the SYM-H index. The vertical dashed red line marks the onset of the storm's main phase. To account for variability in storm duration, the time axis of each individual event is rescaled such that the durations of the initial, main, and recovery phases match their respective mean durations across all events. In this way, each storm is stretched or compressed in time so that phase boundaries are aligned across events. The normalized time axis is expressed in units of hours. Each normalized phase is subdivided into time bins, and parameter values from individual storms are mapped onto these bins for the superposed epoch analysis.

At each normalized time bin, statistical measures including the mean, median, and interquartile range (25th and 75th percentiles) are computed across all contributing events. The mean represents the average behavior, while the median and quartiles provide a robust characterization of the distribution and its variability, allowing us to assess the event-to-event spread that is not captured by the mean alone.

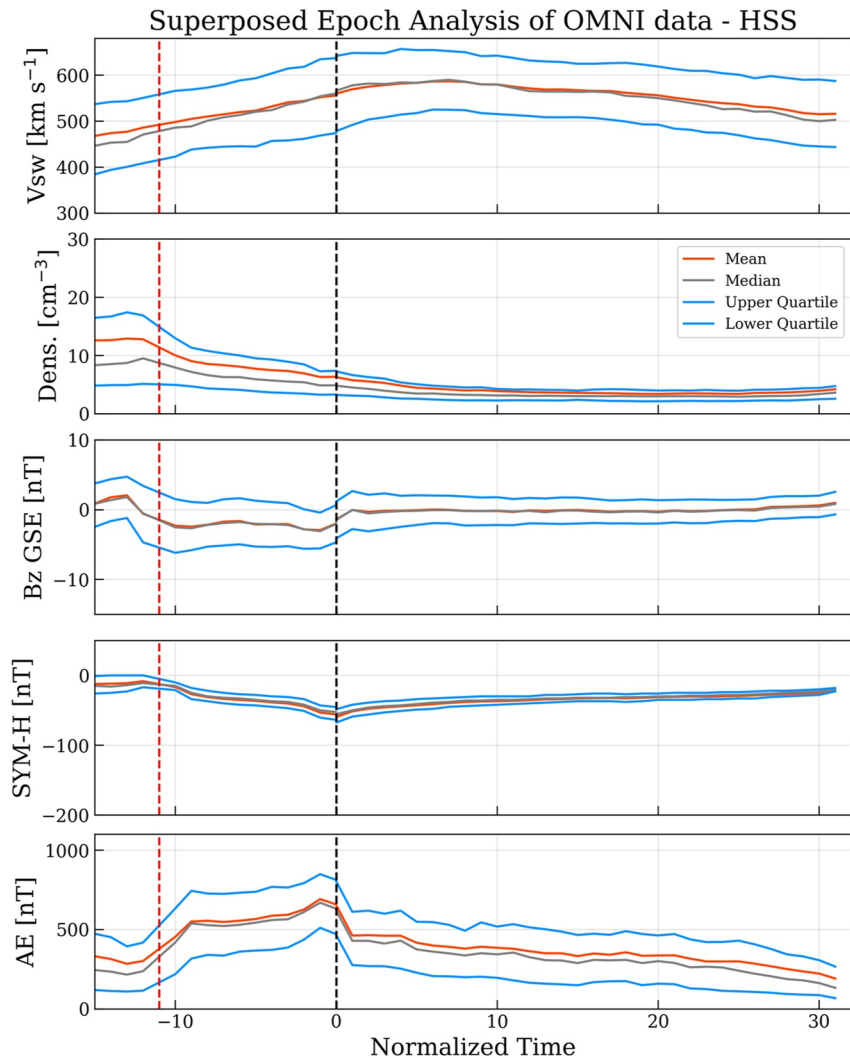


Figure 4. Time-normalized superposed epoch analysis of solar wind and geomagnetic parameters during interplanetary coronal mass ejection (ICME) events. From top to bottom: solar wind speed, proton density, interplanetary magnetic field B_z component, SYM-H index, and auroral electrojet index. The red line represents the mean, the gray line the median, and the blue lines indicate the upper and lower quartiles, illustrating the variability across events. The same y-axis ranges are used for both ICME and high-speed stream panels to enable direct comparison of magnitudes between the two event types.

The period preceding the red dashed line represents the average initial phase, followed by the main phase and the recovery phase. Each figure is organized, from top to bottom, as follows: solar wind speed, proton density, IMF B_z component, SYM-H index, and AE index.

The analysis includes storms spanning a wide range of intensities, from weak to extreme events. While this approach enables a comprehensive characterization of typical responses, it may also introduce variability in the superposed results. However, no single intensity class dominates the statistics. The inclusion of median and interquartile ranges helps to assess the robustness of the observed patterns across different event intensities.

For ICME events (Figure 3), the proton density increases prior to the main phase, indicating the arrival of the sheath region, followed by a decrease associated with the ICME. The solar wind speed remains elevated (around 600 km/s) throughout the main and recovery phases. Immediately following the minimum of the SYM-H index, a minor, transient enhancement in velocity is observed. This feature is consistent with the superposed epoch profiles of ICME-driven storms reported by Kataoka and Miyoshi (2006), reflecting the internal expansion dynamics of ICMEs where the trailing portion maintains higher relative speeds. The IMF B_z turns southward before the main phase, remains negative during it, and recovers toward zero afterward. Correspondingly, the SYM-H

index drops sharply at the onset of the main phase and gradually recovers, while the AE index increases prior to the epoch, peaks near the main phase onset, and remains elevated during recovery.

The SYM-H index shows a sharp and sustained decrease following the start of the main phase, reaching a minimum during the storm's peak. The magnitude of this decrease confirms the occurrence of moderate to intense geomagnetic storms triggered by the ICME events (Gonzalez et al., 1994). The steep initial drop corresponds to the development of the main phase, driven by the enhanced formation of the ring current due to the injection of energetic particles (Ferradas et al., 2023; Fok et al., 1996). During the recovery phase, SYM-H gradually returns to baseline, reflecting the slow decay of the ring current as the geomagnetic system re-equilibrates (Fok et al., 1996; Reyes et al., 2021). The AE index demonstrates significant variability throughout the event. It shows a marked increase before the zero epoch, peaking near the onset of the main phase. This indicates heightened AE activity, a result of enhanced ionospheric convection and substorm intensification driven by the southward IMF. Post-epoch, the AE index remains elevated, indicating ongoing auroral and polar region disturbances, even as the storm subsides. This prolonged activity is typical of ICMEs, which often cause extended intervals of enhanced substorm activity due to sustained coupling between the solar wind and the magnetosphere (Akasofu, 2021).

ICME-driven geomagnetic storms are characterized by their intensity and duration, often resulting in significant impacts on space weather. The elevated solar wind speed, sustained southward IMF B_z , and prolonged AE activity emphasize the complex and multi-faceted nature of geomagnetic responses to ICMEs. These storms not only enhance the ring current but also drive widespread disturbances in the ionosphere and magnetosphere, affecting both high-latitude auroral regions and mid-latitude systems.

For HSS events (Figure 4), the solar wind speed gradually increases, peaking after the epoch and then declining. Proton density shows a modest enhancement before the main phase, without the pronounced sheath signature seen in ICME events. The IMF B_z exhibits weaker and shorter southward excursions, leading to a more gradual and less intense decrease in SYM-H. The AE index shows a moderate increase, with less pronounced and shorter-lived peaks compared to ICME-driven storms.

The solar wind flow speed, V_{SW} , behaves differently between the two types of storm drivers, as we can see in Figures 3 and 4. For ICMEs, the speed remains high, averaging around 600 km/s during the main and recovery phases. In contrast, HSS events exhibit a gradual increase in flow speed, peaking after the epoch, and then slowly declining. This gradual acceleration is typical of high-speed streams, where the fast solar wind overtakes and compresses slower plasma, forming co-rotating interaction regions (CIRs) at their boundaries (Burlaga, 1974; Gosling et al., 1978). The proton density also differs significantly between the two event types. For ICMEs, there is a sharp increase in density before the main phase, coinciding with the arrival of the dense plasma from the ICME's sheath region, followed by a gradual decline. HSS events, on the other hand, show a less pronounced increase in density, peaking slightly before the main phase. This difference reflects the absence of a dense sheath region in HSS events, where density enhancements primarily result from corotating interaction region (CIR) compression effects (Borovsky & Denton, 2006).

The IMF B_z component reveals a pronounced and sustained southward turning before the main phase during ICME events, which is critical for initiating strong geomagnetic activity. This negative B_z persists throughout the main phase before recovering near zero in the recovery phase. Conversely, HSS events display a relatively weaker and shorter duration of southward B_z turning, which explains the typically moderate or weak geomagnetic responses associated with these events. The transient nature of the southward IMF during HSS events limits the extent of magnetic reconnection and the intensity of geomagnetic disturbances.

The SYM-H index shows a sharp and deep decrease during the main phase of ICME events, indicative of strong geomagnetic storms driven by ring current intensification. The index reaches its minimum and then gradually recovers, consistent with the prolonged impact of the ICME on the magnetosphere. For HSS events, the decrease in SYM-H is more gradual and less intense, reflecting the moderate development of the ring current due to the sustained, but weaker, solar wind-magnetosphere coupling. The AE index highlights the differences in auroral activity between the two event types. For ICME events, AE exhibits a sharp increase before the zero epoch, peaking near the main phase onset, and remains elevated for an extended period. This reflects intense and sustained AE activity, driven by prolonged southward IMF conditions and enhanced ionospheric convection. In contrast, HSS events show a more gradual increase in AE, with a peak that is less pronounced and shorter in

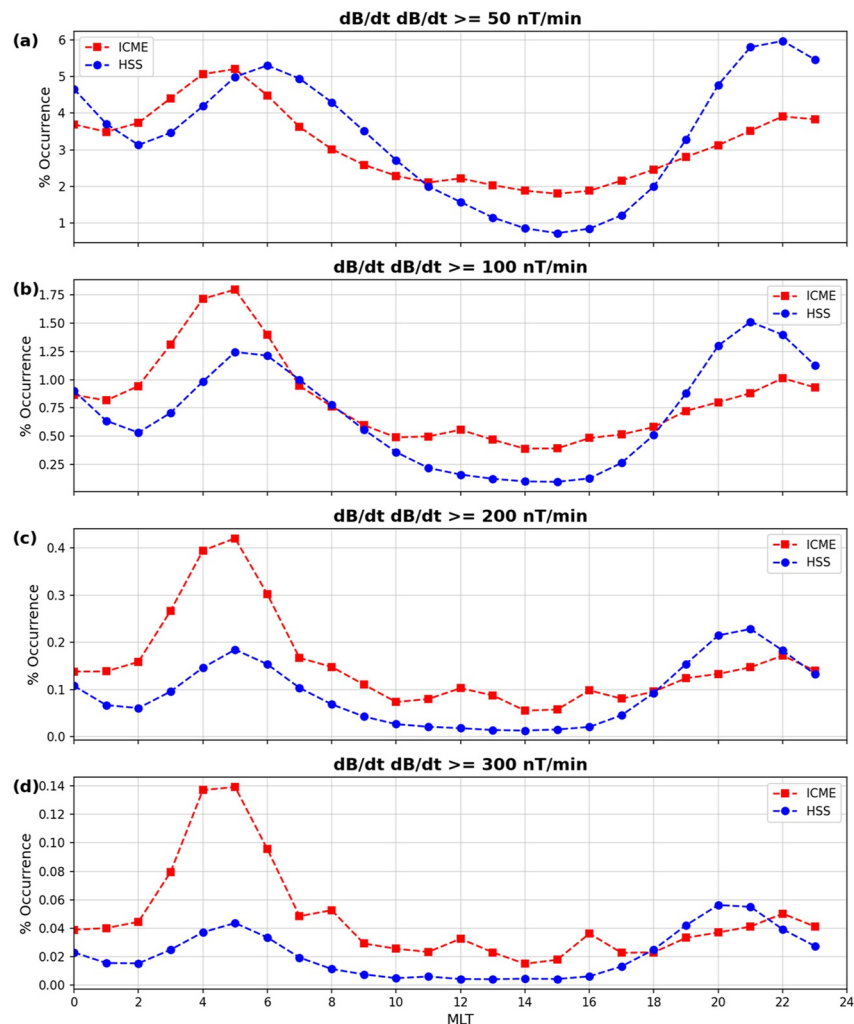


Figure 5. Distribution of dB_H/dt spikes across magnetic local time (MLT) sectors for different spike magnitude thresholds. Panels (a–d) show the number of spikes per MLT sector for HSS-related events (blue circles) and ICME-related events (red squares): (a) $dB_H/dt \geq 50$ nT/min, (b) $dB_H/dt \geq 100$ nT/min, (c) $dB_H/dt \geq 200$ nT/min, and (d) $dB_H/dt \geq 300$ nT/min.

duration. The post-epoch AE values for HSS events are relatively moderate, suggesting less persistent auroral disturbances than ICME-driven storms.

Overall, ICMEs generate more intense geomagnetic storms due to the combined effects of elevated solar wind speeds, dense plasma, and sustained southward IMF conditions. In contrast, HSS events typically lead to moderate geomagnetic activity, characterized by gradual changes in solar wind parameters and recovery phases that return to quiet levels slowly. These distinct parameter profiles during ICME and HSS events are consistent with classical superposed epoch analyses established in earlier literature (Kataoka & Miyoshi, 2006). The distinct storm profiles associated with these event types have important implications for space weather forecasting and impact assessment on technological systems (Alves et al., 2011; Borovsky & Denton, 2006; Zhang et al., 2007).

Figure 5 presents the occurrence distribution of dB_H/dt spikes as a function of Magnetic Local Time (MLT) for both ICME- and HSS-driven events. The panels represent cumulative thresholds of spike magnitude: (a) >50 nT/min, (b) >100 nT/min, (c) >200 nT/min, and (d) >300 nT/min. Because these thresholds are inclusive, each panel contains all spikes exceeding the stated value, meaning higher-magnitude spikes shown in panel (d) are also accounted for in the distributions of panels (a) through (c).

Across all thresholds, the distribution of spikes is not uniform in MLT. For both drivers, two prominent peaks are observed: one in the pre-midnight sector (around 22 MLT) and another in the dawn sector (around 5 MLT). These features are most prominent at lower thresholds (panel a) and remain visible, though less pronounced for HSS-driven events, at higher thresholds (panels b–d). For HSS-driven events, the pre-midnight peak is present at all thresholds, while the dawnside peak is less pronounced. The dawnside peak for HSS-driven events is shifted noonward compared to the same peak for ICME-driven storms. This phenomenon is particularly visible at lower thresholds. As the dB_H/dt threshold increases from panel (a) to (d), the spikes decrease for both drivers, and the distributions become more concentrated around the pre-midnight sector. The comparison between ICME and HSS events shows that ICME-driven spikes are more frequent and more likely to reach higher magnitudes, particularly in the dawn sector. This suggests that the processes responsible for large dB_H/dt variations are more efficiently triggered by ICME conditions, especially in the dawn and pre-midnight MLT sectors. For HSSs, high-magnitude spikes are lower and more confined to the pre-midnight sector.

Previous work by Milan et al. (2023) identified three MLT regions with enhanced dB/dt activity: pre-midnight (linked to substorm onsets), post-midnight (associated with omega band activity), and pre-noon (related to Kelvin-Helmholtz Instabilities, KHI, at the magnetopause). Their analysis, which did not distinguish between solar wind drivers, showed that substorm-related spikes are more common during moderate and weak solar wind activity, omega bands during strong solar wind, and KHI during quiet conditions but with high solar wind speeds ($>600 \text{ km s}^{-1}$).

Our observations of pre-midnight and dawn sector peaks are consistent with earlier statistical studies of dB/dt activity. For instance, Viljanen and Tanskanen (2011) found that intense dB_H/dt values in northern Europe were most pronounced around midnight and early morning hours, diminishing significantly by noon and early afternoon. Similarly, Schillings et al. (2022), utilizing global magnetometer networks, showed that magnetic field spikes during geomagnetic storms exhibit distinct behavior compared to non-storm periods, with clear enhancements in the pre-midnight and morning sectors. Our work further refines this understanding by demonstrating that the relative contributions and characteristics of these peaks vary significantly based on the solar wind driver.

The strong pre-midnight concentration observed across all dB_H/dt thresholds for both ICME and HSS events, and particularly evident for HSS-driven events, strongly suggests the dominant role of substorm expansion phase intensifications and the associated westward AE (Akasofu, 1964). During substorms, large and rapid changes in the westward electrojet current system lead to significant horizontal magnetic field perturbations, which are efficiently captured as dB_H/dt spikes. The fact that this peak is prominent even for higher thresholds and in HSS events (which generally cause weaker, prolonged storms than ICMEs) underscores the fundamental role of substorm dynamics in generating hazardous rapid geomagnetic changes (Juusola et al., 2015). This suggests that even moderate solar wind coupling from HSS can effectively trigger substorms, leading to localized but intense dB_H/dt enhancements on the nightside.

Conversely, the more pronounced dawn-sector peak, particularly for ICME-driven events and especially for larger spikes ($>100 \text{ nT/min}$), points to an enhanced contribution from phenomena like omega bands (Milan et al., 2023, 2024). Omega bands are large-scale, undulating structures of the AE that propagate eastward in the post-midnight to dawn sector, often associated with strong geomagnetic activity and enhanced eastward electrojet currents. The higher likelihood of these events during ICME-driven storms, which are characterized by more intense and dynamic solar wind conditions (e.g., strong southward IMF B_Z), suggests that ICMEs are more effective at driving the magnetospheric conditions conducive to omega band formation and the associated large dB_H/dt fluctuations in the dawn region. This distinction highlights that while substorms are a ubiquitous source of dB_H/dt for both drivers, other processes like omega bands become particularly significant during the more intense forcing provided by ICMEs.

While less dominant than the nightside peaks, some dB_H/dt activity is still observed in the dayside sectors (around pre-noon and post-noon MLT) for both ICME and HSS events, though at generally lower occurrence rates and magnitudes. This dayside activity could be related to various mechanisms, including the generation of Ultra-Low Frequency (ULF) waves such as Pc5 pulsations. These pulsations can be excited by Kelvin-Helmholtz Instabilities (KHI) at the magnetopause, particularly during periods of high solar wind velocity. While Milan et al. (2023) associated KHI with high solar wind speeds even during quiet conditions, our observation of some dayside activity during both ICME and HSS events (which often involve elevated solar wind speeds) suggests that

these mechanisms can contribute to the overall dB_H/dt budget, albeit to a lesser extent than the nightside processes. Other dayside sources might include sudden impulses (SIs) from solar wind dynamic pressure enhancements, particularly during ICME sheath passages.

The sensitivity of dB_H/dt MLT distribution and magnitude to the solar wind driver underscores the complex interplay between solar wind forcing and magnetospheric response. ICME-driven events, characterized by abrupt and intense changes in solar wind parameters (e.g., sustained southward IMF B_Z , high dynamic pressure), appear to drive more widespread and higher-magnitude dB_H/dt spikes across both the pre-midnight and dawn sectors, likely by activating a broader range of current systems and instabilities. In contrast, HSS-driven events, while often leading to prolonged geomagnetic activity, tend to produce dB_H/dt spikes that are more consistently concentrated in the pre-midnight sector and generally of lower maximum magnitude, reflecting a more dominant role for substorm activity. This differentiation is critical for improving space weather forecasting models and for targeted mitigation efforts against GICs, as the specific hazard profile (location, intensity, duration) varies significantly with the solar wind driver.

Our results indicate that the MLT distribution and magnitude of dB_H/dt spikes are sensitive to both solar wind structure and the intensity of geomagnetic activity. We demonstrate that by distinguishing between ICME and HSS drivers and by considering different dB_H/dt thresholds. ICME-driven events favor both dawn and pre-midnight sectors, especially for larger spikes, while HSS-driven events show a less pronounced dawnside peak and fewer high-magnitude spikes. This distinction provides further insight into the magnetospheric processes responsible for rapid geomagnetic field variations under different solar wind conditions.

4. MLT and Latitude dB_H/dt Distribution

Figure 6 shows the distribution of dB_H/dt spikes as a function of MLT for stations located above 50° magnetic latitude in the Northern Hemisphere. Each panel presents the distribution (expressed as a percentage) of spike events within 1-hr MLT and 3° magnetic latitude bins, with noon at the top and midnight at the bottom. Concentric circles indicate magnetic latitude, ranging from 90° to 50° . The color scale represents the percentage of spikes computed relative to the total number of spikes occurring within each storm phase (initial, main, or recovery) and for each solar wind driver type (ICME or HSS). The left column displays the distributions for ICME-related events, while the right column shows those associated with HSSs. The top, middle, and bottom rows correspond to the initial, main, and recovery phases of the geomagnetic storms, respectively.

During the initial phase of the ICME-related storms, the plots show a relatively dispersed distribution of dB_H/dt spikes across various MLT hours, indicating that the storm's onset impacts a broad MLT range. While the distribution is widespread, the relative percentage of spikes is notably higher in the dawn sector compared to other MLT sectors, with a moderate presence over the dayside. The widespread distribution during the initial phase is consistent with the sudden compression of the magnetosphere by the ICME sheath region, which can generate SIs and SSCs that propagate globally and induce dB_H/dt across a broad region. This global compression also triggers localized enhancements in the post-midnight sector at higher latitudes, likely due to rapid tail stretching and early nightside current system activations prior to the main phase onset. The higher count in the dawn sector during this phase may be influenced by localized enhancements resulting from early plasma injections or pre-existing dawnside current systems interacting with the sudden pressure increase.

As ICME-driven storms progress into the main phase, Figure 6c conspicuously shows a higher concentration of dB_H/dt spikes in the post-midnight sector (around 03–06 MLT) at auroral latitudes (60° – 70° MLAT). This pronounced clustering of spike occurrences is characteristic of the main phase of intense storms, driven by the sustained southward IMF B_Z component within the ICME, leading to enhanced magnetic reconnection, energy injection into the magnetosphere, and intense AE activity. The post-midnight hotspot specifically aligns with regions known for the generation and propagation of omega bands and other dynamic auroral phenomena (e.g., pulsating aurora, Pc5 pulsations) that can cause rapid magnetic field variations. The recovery phase of ICME storms (Figure 6e) still exhibits activity in both pre- and post-midnight sectors, though less localized and with lower occurrence than during the main phase. This behavior reflects the gradual decay of the ring current and the continued influence of multiple magnetospheric and ionospheric processes, rather than being attributable to a single mechanism (Akasofu, 2021; Daglis et al., 1999; Fok et al., 2001).

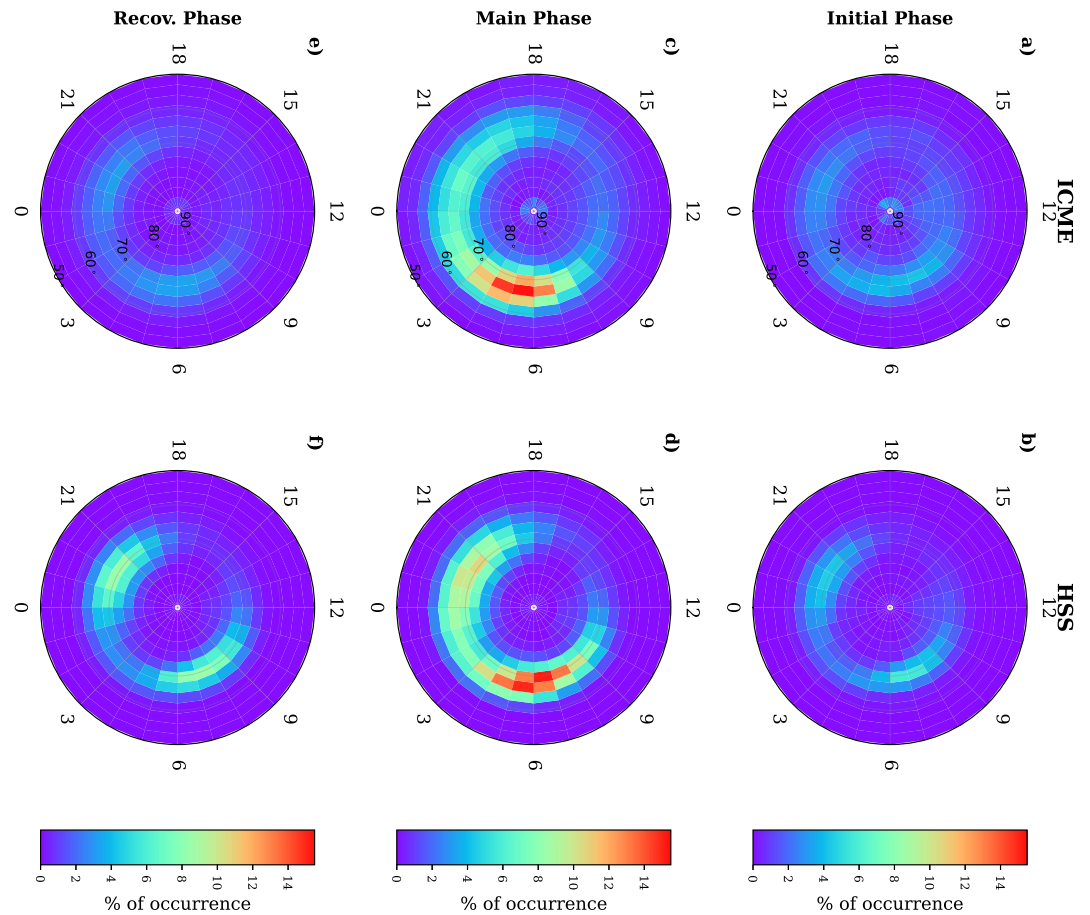


Figure 6. Distribution of dB_H/dt spikes in magnetic local time and latitude, from 50° to 90° in both hemispheres.

For HSS-driven events, the dB_H/dt spikes exhibit distinct patterns as seen in Figures 6b–6d and 6f. Unlike ICMEs, a predominant pre-midnight hotspot is observed across all storm phases (initial, main, and recovery), particularly concentrated between 60° and 70° MLAT. Nevertheless, a secondary, less intense enhancement remains visible in the post-midnight sector across all these phases. While the main phase of HSS events (Figure 6d) shows a more defined secondary enhancement in the post-midnight sector, the pre-midnight occurrence remains higher. This consistent pre-midnight preference for HSS-related spikes strongly suggests that substorm onset and the intensification of the westward AE are the primary drivers of rapid geomagnetic variations during these events. In contrast, storms associated with HSSs are initiated by the compression of Corotating Interaction Regions and sustained by the subsequent fast wind. This combined forcing efficiently drives recurrent substorm activity over a prolonged period, lacking the extreme initial compressions often observed in ICME events.

Comparing the two drivers, ICME-driven events demonstrate higher relative occurrence of dB_H/dt spikes. This is particularly evident during the main phase in the post-midnight sector, where the percentage of occurrence reaches up to almost 20% bins. In contrast, HSS events show lower maximum percentages that are primarily confined to the pre-midnight sector. Additionally, the dawn-side peak occurs closer to noon for HSS events than for ICMEs, which aligns with our earlier observations (Figure 5) showing that ICME-driven spikes are more frequent and reach higher magnitudes in the dawn sector. These differences support the idea that ICMEs, with their intense and turbulent solar wind conditions, trigger a broader range of magnetospheric processes, such as omega bands (Engelbreton et al., 2020; Forsyth et al., 2020; Milan et al., 2023), that drive large dB_H/dt values across the entire nightside. Conversely, HSS events are more consistently dominated by substorm-related processes in the pre-midnight sector. Ultimately, these distinct MLT and latitudinal distributions underscore how each solar wind structure uniquely couples with the magnetosphere, creating separate hazard profiles for ground-based technological systems.

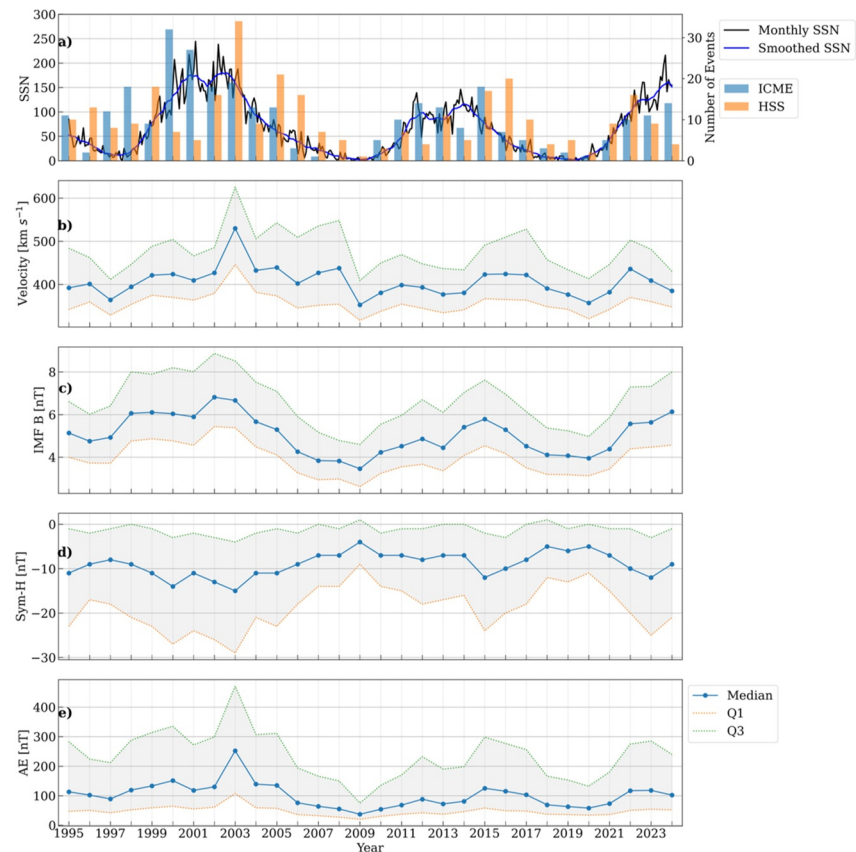


Figure 7. Solar cycle variation of driver-specific storms, solar wind parameters, and geomagnetic activity indices from 1995 to 2024. Panel (a) shows the monthly sunspot number (SSN) as a black line and its smoothed trend in blue, along with the annual counts of ICME-driven storms (blue bars) and HSS-driven storms (orange bars). Panel (b) presents the median solar wind velocity (blue line) with the interquartile range (Q1 and Q3 shown as orange and green dotted lines, respectively) for interplanetary coronal mass ejection (ICME) and high-speed stream (HSS) events. Panel (c) displays the median interplanetary magnetic field magnitude with corresponding quartiles. Panel (d) shows the median SYM-H geomagnetic index, indicating ring current strength, with quartile ranges. Panel (e) illustrates the median auroral electrojet index, a measure of auroral activity, with its interquartile range. The shaded areas in panels (b–e) highlight the variability of these parameters over the solar cycle for both ICME and HSS events. This figure highlights the solar cycle dependence of solar wind drivers and their geomagnetic impacts.

5. Solar Cycle Dependence of dB_H/dt

Figure 7 illustrates the solar cycle dependence of key solar wind and geomagnetic parameters by comparing their annual variations with the sunspot number (SSN). Panel (a) presents the yearly counts of solar wind events, distinguishing between ICMEs (blue bars) and HSSs (orange bars), alongside the monthly averaged and smoothed SSN (black and blue lines, respectively). Panel (b) displays the annual distribution of solar wind velocity, highlighting the median values and the interquartile range (25th and 75th percentiles). Panel (c) shows the annual distribution of the IMF magnitude, presented in the same format as panel (b), with median and interquartile ranges. Panels (d) and (e) show the SYM-H and AE index, respectively. This figure demonstrates a dependence of solar wind conditions on the solar cycle, particularly in the occurrence of ICME-driven and HSS-driven storms. The temporal distribution of these geomagnetic storms follows the expected pattern dictated by the solar cycle, with ICME-driven events being more frequent during the ascending and peak phases of the cycle. In contrast, HSS-driven storms are predominantly concentrated in the cycle's declining phase. This behavior aligns with well-established solar-terrestrial coupling mechanisms, in which distinct solar wind drivers characterize different phases of the solar cycle (Borovsky & Denton, 2006; Denton et al., 2006; Yermolaev et al., 2012). Furthermore, our long-term solar cycle distribution shows great consistency with the multi-cycle statistical analysis presented

by Hajra et al. (2021), reinforcing how these driver-specific occurrence rates and typical geomagnetic responses persist across consecutive solar cycles.

The occurrence of ICME-driven storms is notably higher during the maximum and late ascending phases of Solar Cycles 23 and 24, coinciding with periods of enhanced solar activity. This is consistent with the increased frequency of coronal mass ejections (CMEs) from the Sun during these phases, leading to a greater number of interplanetary manifestations reaching Earth's vicinity. A comparison between the two cycles reveals that the ICME count was higher during Cycle 23, indicating a stronger overall activity level for this cycle compared to Cycle 24. This finding aligns with previous studies, which indicate that Cycle 24 exhibited a lower ICME rate and overall weaker solar activity, ultimately influencing the number of ICMEs observed in interplanetary space (Kilpua et al., 2015; Shen et al., 2017). Furthermore, as demonstrated by Wang and Colaninno (2014) and Gopalswamy et al. (2015), even though the weakened heliospheric pressure in Cycle 24 allowed a high volume of ICME expansions, their resulting magnetic structures at 1 AU possessed weaker core magnetic fields, leading to a general reduction in geoeffectiveness and a lower count of storm-triggering ICMEs compared to Cycle 23 (Rawat et al., 2018).

The behavior of high-speed streams (HSS) during the declining phase of the solar cycle provides a more continuous and structured representation of solar cycle evolution compared to transient events such as ICMEs. Unlike ICMEs, which are essentially stochastic and dependent on the occurrence of active regions, HSS events follow a more predictable pattern dictated by the long-term evolution of coronal holes. These persistent structures, which become more prominent during the declining phase of the solar cycle, serve as stable sources of enhanced solar wind speeds, producing recurrent geomagnetic activity as they interact with Earth's magnetosphere (Collin et al., 2025; Mursula et al., 2015).

The influence of HSS on geomagnetic activity is also evident in the occurrence of rapid geomagnetic variations, particularly those characterized by high values of dB_H/dt . While large dB_H/dt fluctuations are often linked to SSCs triggered by ICMEs (Kappenman, 2003; Nilam & Tulasi Ram, 2022), it is increasingly evident that HSS events also contribute significantly to rapid geomagnetic field variations. This can be attributed to the formation of CIRs at the leading edge of HSS, where fast solar wind encounters preceding slow wind, leading to enhanced geoeffectiveness (Grandin et al., 2019). The compression regions formed in these interactions generate prolonged periods of moderate geomagnetic disturbances, which, although typically less intense than those driven by ICMEs, can still produce substantial dB_H/dt fluctuations. The temporal distribution of dB_H/dt occurrences follows the pattern of HSS prevalence, particularly during the declining phase of the solar cycle (Grandin et al., 2019). This highlights the necessity of considering HSS as a key driver in space weather impact assessments, particularly concerning the risks posed by GICs. Many studies have traditionally emphasized ICME-driven disturbances due to their immediate and often severe impacts. However, the recurrent nature of HSS-driven geomagnetic activity suggests that their cumulative effects should not be overlooked. Although HSS-driven events generally exhibit lower peak amplitudes than ICMEs, their prolonged duration drives sustained, low-level GICs. Over several days, this continuous activity can lead to progressive thermal stress and saturation in power grid transformers, as well as accelerated corrosion in long pipeline networks. This underscores the importance of incorporating HSS dynamics into predictive models of geomagnetic variability and space weather forecasting.

Figure 8 presents the percentage distributions of dB_H/dt occurrences during selected solar minima and maxima periods, segmented by geomagnetic storm phases (initial, main, and recovery). Panels on the left correspond to solar maxima (2000–2002, 2013–2015, and 2022–2024), while panels on the right represent solar minima (1995–1997, 2007–2009, and 2018–2020). To capture the current solar cycle within our 29-year database, the 2022–2024 interval was included in the solar maximum category, effectively encompassing the strong ascending phase and peak activity of Solar Cycle 25 available up to the limit of our data coverage. Each panel displays the relative frequency of dB_H/dt events, with prominent peaks around 12%–18% in the main and recovery phases during both cycle stages, though with notable differences in distribution shapes and maxima. During solar minima, the distributions reveal a higher prevalence of dB_H/dt events in the recovery phase (panel f), compared to the initial (panel b) and main (panel d) phases, which show more subdued peaks around 15%–16%. This pattern suggests that prolonged, moderate disturbances, often associated with HSS and CIRs, dominate geomagnetic variability in quieter solar periods, leading to elevated dB_H/dt risks during storm recovery. In contrast, solar maxima exhibit broader and more intense distributions across all phases (panels a, c, e), with the main phase

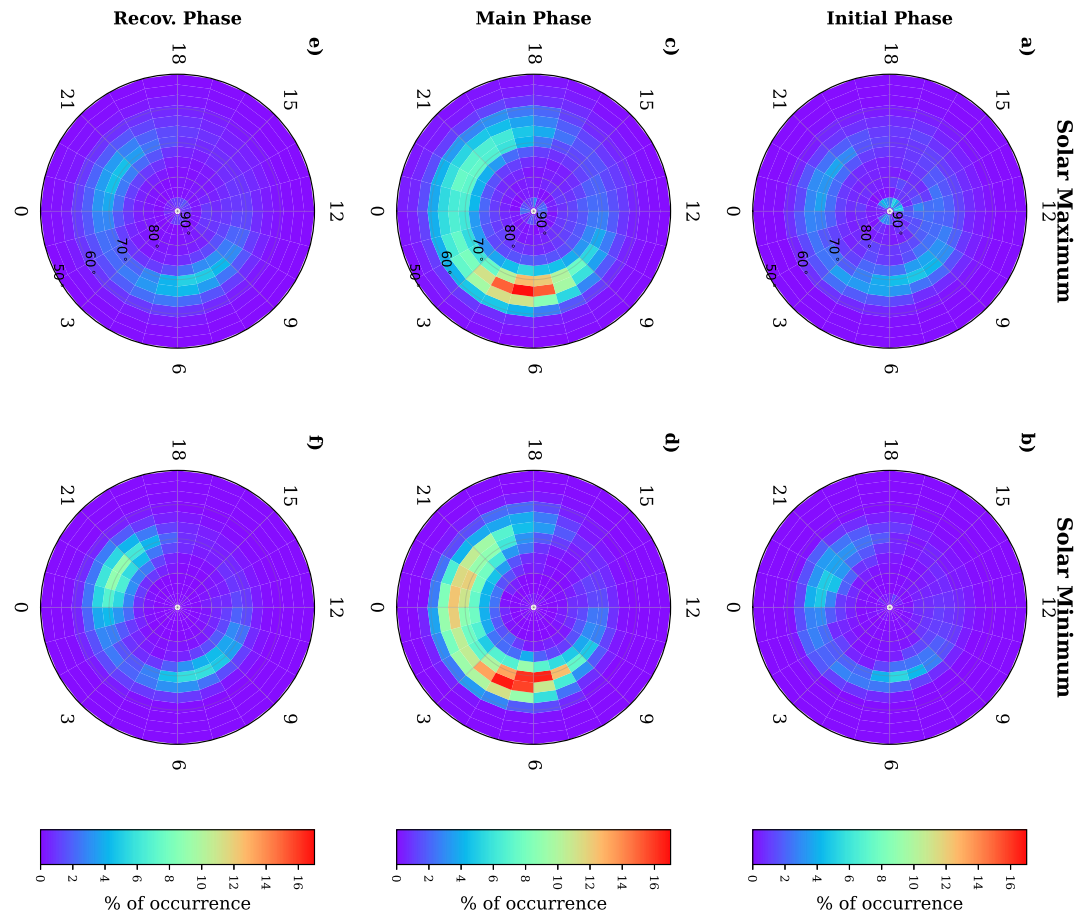


Figure 8. Distribution of dB_H/dt occurrences during solar maxima (left panels: a, c, e) and solar minima (right panels: b, d, f), categorized by geomagnetic storm phases: initial phase (a, b), main phase (c, d), and recovery phase (e, f). Data cover solar minima (1995–1997, 2007–2009, 2018–2020) and maxima (2000–2002, 2013–2015, 2022–2024).

(panel c) showing the highest occurrences (up to 21%), indicative of ICME-driven sudden commencements and intense substorms. The initial phase during maxima (panel a) also displays elevated frequencies (e.g., 16%–18%), aligning with the increased frequency of ICME occurrences that triggers rapid field changes at storm onset.

These differences underscore the solar cycle's modulation of dB_H/dt drivers: ICMEs contribute to sharper, more frequent peaks during maxima, while HSS foster sustained but less extreme variations during minima. This cycle-dependent behavior has implications for space weather forecasting, as models must account for phase-specific vulnerabilities—such as heightened recovery-phase risks in minima—to mitigate infrastructure impacts. Future analyses could extend this by incorporating latitudinal variations or direct GIC measurements to refine predictive frameworks.

6. Discussion

The MLT distribution of dB_H/dt spikes reveals clear and systematic differences between ICME- and HSS-driven events, reflecting their distinct magnetospheric coupling mechanisms. For HSS-related perturbations, spikes predominantly concentrate in the pre-midnight sector across all storm phases (initial, main, and recovery), indicating the dominant role of substorm-related processes and westward electrojet intensifications. In contrast, ICME-related spikes exhibit a stronger post-midnight preference during the main phase, together with a broader distribution extending across multiple MLT sectors during the initial and recovery phases. These results demonstrate that the spatial organization of rapid geomagnetic field variations is strongly controlled by the solar wind driver and the associated energy transfer pathways.

Previous studies have identified key MLT “hotspots” for geomagnetic perturbations by treating events as a single population, without explicitly distinguishing between different solar wind drivers (A. Fleetham et al., 2024; L. A. Fleetham, 2024; Milan et al., 2023; Schillings et al., 2022). For instance, Milan et al. (2023) highlighted three primary regions: pre-midnight (linked to substorm onsets), dawn (associated with omega band activity), and pre-noon (related to Kelvin-Helmholtz instabilities at the magnetopause). Complementary analyses have further demonstrated that rapid magnetic field variations are closely associated with magnetosphere-ionosphere coupling processes, particularly FACs and substorm dynamics, with spikes frequently co-located with regions of enhanced current systems (A. Fleetham et al., 2024).

Our results extend these findings by showing that these mechanisms are not uniformly active, but instead depend strongly on the solar wind driver. In particular, the pre-midnight hotspot dominates during HSS-driven storms, consistent with substorm-driven westward electrojet intensifications (Schillings et al., 2022), whereas ICME-driven storms exhibit enhanced post-midnight activity during the main phase, associated with omega band dynamics and intensified current systems. Similarly, Viljanen and Tanskanen (2011) reported pronounced dB_H/dt activity around midnight and early morning. Here, we refine this picture by demonstrating that pre-midnight dominance is primarily characteristic of HSS-driven moderate storms, while ICME-driven intense storms activate multiple current systems across a broader MLT range, consistent with stronger and more complex magnetosphere-ionosphere coupling.

The intensity and phase-dependent distribution of dB_H/dt spikes further distinguish ICME and HSS events. ICME-driven spikes reach peak intensities during the initial and main phases and decrease during recovery, consistent with the presence of turbulent sheath regions and sustained southward IMF B_Z conditions that enhance rapid magnetic field variability (Huttunen et al., 2008). In contrast, HSS-driven spikes maintain more uniform but generally lower intensities across all storm phases, reflecting prolonged and moderate solar wind forcing associated with co-rotating interaction regions (CIRs). This behavior is consistent with the classification of geomagnetic disturbances proposed by Kataoka and Ngwira (2016), where ICME-driven events are associated with impulsive and strongly driven processes, while HSS-driven storms are linked to more gradual, convection-dominated dynamics. Furthermore, recent studies have shown that within these long-duration events, the highly compressed fields of CIR-driven phases can produce intense localized GIC bursts. This is due to enhanced wave activity and fine-scale structuring (Pilipenko et al., 2024), highlighting the importance of mesoscale processes in shaping extreme dB_H/dt events.

These driver-dependent differences are further modulated by the solar cycle. ICME-driven events contribute to sharper and more frequent dB_H/dt peaks during solar maximum, reflecting the increased occurrence of ICMEs and stronger solar wind forcing. In contrast, HSS-driven activity dominates during the declining and minimum phases of the solar cycle, producing sustained but less extreme geomagnetic variability (Juusola et al., 2023; Zhang et al., 2021). Crucially, our long-term data set incorporates the recent 2022–2024 interval, capturing the strong ascending phase and peak activity plateau of Solar Cycle 25. The inclusion of these recent extreme intervals confirms that the distinct driver-specific hazard profiles observed in previous decades remain highly predictable and persistent across consecutive solar cycles. This cycle-dependent behavior underscores the importance of long-term variability in geomagnetic hazard assessment. While ICME-driven storms pose a higher risk of short-lived extreme disturbances, HSS-driven activity can lead to prolonged exposure to moderate perturbations. This continuous activity potentially results in cumulative impacts on technological systems, such as the accelerated aging of power grid components, gradual transformer degradation, and increased long-term maintenance costs for ground-based infrastructure.

Overall, our results demonstrate that the occurrence and characteristics of dB_H/dt spikes are controlled by the combined effects of solar wind driver type, storm phase, and solar cycle variability. By explicitly separating ICME- and HSS-driven events and analyzing their behavior over nearly three decades, this study provides a unified framework that extends previous work by linking magnetospheric processes, spatial distributions, and long-term variability. These findings have direct implications for space weather forecasting and risk assessment, particularly in improving our ability to predict when and where hazardous geomagnetic perturbations are most likely to occur.

7. Conclusions

In this study, we investigated the spatiotemporal characteristics and long-term solar cycle dependence of rapid geomagnetic field variations (dB_H/dt) associated with ICME- and HSS-driven storms. By leveraging a comprehensive 29-year high-resolution database (1995–2024) that encompasses the peak of Solar Cycle 25, we established a phase-resolved statistical framework of ground magnetic perturbations.

The main findings of this work can be summarized as follows:

1. The high-resolution MLT distribution of dB_H/dt spikes differs significantly between solar wind drivers. HSS-driven events are strictly dominated by a localized pre-midnight hotspot across all phases, whereas ICME-driven events activate broader nightside and dawn current systems, exhibiting an enhanced post-midnight concentration specifically during the storm main phase.
2. ICME-driven storms produce more intense and impulsive dB_H/dt spikes, with hazard rates heavily concentrated during the initial and main phases. Conversely, HSS-driven storms generate sustained, long-duration risks that remain remarkably elevated even through the prolonged recovery phase.
3. The observed spatiotemporal variability reflects distinct magnetospheric coupling timescales, where ICME perturbations are driven by impulsive energy injections and compressed sheath regions, while HSS perturbations are maintained by recurrent, Alfvénic convection-driven dynamics.
4. The statistical distributions, evaluated through comprehensive median and interquartile ranges, reveal substantial event-to-event variability, demonstrating that traditional mean behavior alone severely underestimates the magnitude and spatial extent of severe geomagnetic spikes.
5. The occurrence, intensity, and phase-specific footprint of dB_H/dt spikes are heavily modulated by the solar cycle, with ICME hazards dominating during solar maxima (including the current Cycle 25) and HSS risks persisting as the primary driver during the declining and minimum phases.

These results demonstrate that geomagnetic hazard is governed not only by bulk storm intensity but also by solar wind driver type, temporal evolution, and local time dependence. This unified phase-resolved approach provides critical refinements for space weather forecasting and for assessing the risks of GICs on vulnerable technological infrastructure across different stages of the solar cycle.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

All data supporting the conclusions of this study are publicly available. Solar wind and interplanetary magnetic field data were obtained from OmniWeb (Papitashvili & King, 2020), available at <https://cdaweb.gsfc.nasa.gov/pub/data/>. Ground-based geomagnetic observations were accessed through the SuperMAG database (SuperMAG, 2012), available at <https://supermag.jhuapl.edu/>. The geomagnetic event catalog developed in this study is openly available via Zenodo (Marchezi & Pinto, 2025) at <https://doi.org/10.5281/zenodo.17967225>. All data processing and analysis were performed using the Python programming language (Python Software Foundation, 2023), with extensive use of the scientific computing libraries NumPy (Harris et al., 2020), SciPy (Virtanen et al., 2020), and pandas (McKinney, 2010). Figures were generated using Matplotlib (Hunter, 2007), an open-source plotting library available at <https://matplotlib.org/>. All software tools used in this study are open source and widely available.

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